PROJECT MANAGEMENT REPORT

Student Name	Student Number
Anastasia Bholee	M01015673
Ivan Pavlovskii	M00957512
Leo D'Hotman de Villiers	M00694663
Terry Leung Chew	M00983500
Ore Orekoya	M00982779

Professor's Name: Gina Soopramanien

Course: 2025-26 CST2550 Software Engineering Management and Development

Submission Date : 14/04/2026

Word Count : 2067

Table of Contents

Table of Figures

1) Project Description

First, our goal was to create a website based on a theme in twelve weeks. The theme assigned to our group was: Bus Timetable and Ticketing System. In our program, coded with WinForms, we decided to create different functionalities for different types of users : the default user would be able to see a bus timetable and buy bus trips to different destinations, the admins would be able to modify and create bus trips, and the moderator would be able to turn an user account into an admin account. To continue, the other requirements consisted of choosing a version control system to upload our website and commit the different changes we made to it; designing and implementing classes to store the data; choosing, designing, implementing and testing a data structure; implementing the program to write and read to a SQL database; and finally implementing an user interface.

The different functionalities and the different requirements will be discussed in more detail later in the report.

2) Team Roles and Responsibilities

Our team consisting of five members, each assigned a specific role, including a Scrum Master, a secretary, two Developers, and a Tester. The Scrum Master serves as the leader, providing structure and guidance to the team while also acting as a mentor in applying Scrum practices. The Secretary supports the Scrum Master by providing organisational and administrative assistance, as well as coordinating team activities such as stand-up meetings to ensure smooth workflow and communication. In addition, the two Developers are responsible for designing and building the product, as well as deciding how items from the backlog can be transformed into a working system. They work closely together to ensure efficiency and reduce miscommunication. Lastly, the Tester ensures that the product is of high quality by creating test cases, identifying bugs, and providing feedback to help the developers improve the overall system.

Each role was assigned based on the strengths and abilities of each team member. The Scrum Master role was assigned to Leo D'Hotman De Villiers due to his strong leadership skills and experience in supporting and uniting team members in group projects. The Secretary role was assigned to Terry Leung Chew because of his strong time management and organisational skills, which helped in scheduling meetings and maintaining clear communication, especially in close coordination with the Scrum Master. The Developer roles were assigned to Anastasia Bholee and Ivan Pavlovskii due to their strong working relationship, which created good synergy and allowed for efficient collaboration with minimal miscommunication. The Tester role was assigned to Ore Orekoya due to his critical thinking skills and attention to detail, which helped in identifying issues and providing effective feedback to improve the product.

Accountability was maintained through consistent standup meetings, as to keep track of the project progress, and to provide feedback from every team member present. Deadlines were set for each feature of the project, as each were strictly monitored though realistic deadlines and timelines as to maintain a consistent workflow for every team member's own individual tasks.

3) Tools and Collaboration Platforms

For our communication platform, Zoom was initially used, but due to the meeting duration of Zoom meetings being heavily limited to 30 minutes, we switched to a platform called Google Meet. As for documentation, after each meeting, we will all write down in a WhatsApp group chat to answer any queries and go through a certain list of questions on what we achieved in the stand-up meetings for each group member. Which would then be recorded using the daily stand-up meeting form which was provided to the group.

As far as maintaining communication between other members, the use of our group chat on WhatsApp was the primary source of communication between each group member. This made it easy to share information which includes documents, code and general ideas in case someone was not able to join the weekly meeting. While also saving any piece of information within the history of the chat. Furthermore, the sharing of information was usually done during the meetings, which helped when it came to instructions for the README files, and allow us to give regular feedback on the code documentations, and tester's report. The meeting notes would also be recorded in a word document, as during the standup meetings, the secretary would feel a form as the meeting goes, to record who was present, and what each member of the team achieved.

Next, concerning the version control system, we chose GitHub because every team member was familiar with this system and found it simpler than Azure DevOps.

4) Meeting Management and Communication

To maintain regular communications in the team and track our progress, we did a meeting each week on Wednesday during the whole duration of the project where each team member talked about the progress of his current task.

After each meeting, Terry, our secretary, would fill out one Daily Stand Up meeting form. Decisions, the accountability of each team member, and the advancement of our project are describe in these documents.

Also, as mentioned above, a WhatsApp group was used if ever a team member needed to share information, ask for help or communicate with the team outside of the weekly meetings.

5) Workload Division

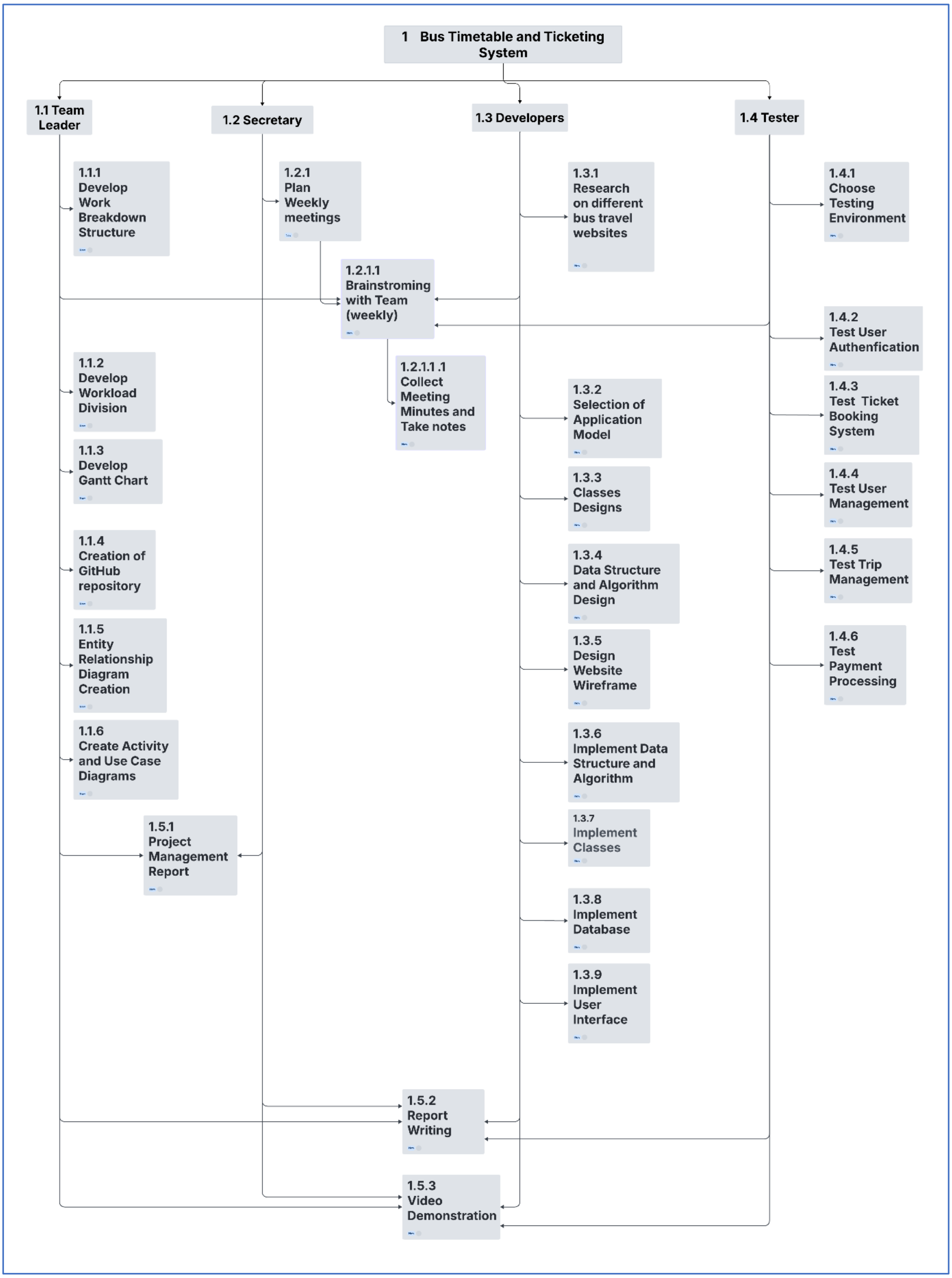

Figure 1 : Workload Division

In Figure 1 above, we can see that all tasks were distributed amongst our team based on their role. Leo, the team leader (Scrum Master), had tasks like developing the Work Breakdown Structure and the Workload Division which are management tasks; he was also responsible for ensuring the synergy and coordination of the team as well as removing impediments. Terry, the secretary, helped Leo with the management, so he was assigned tasks like planning the weekly meetings and taking notes during and after meetings. Anastasia and Ivan, the developers, were assigned all tasks that were related to the code, they made the website. If ever the developers needed help with the code or with the designs, they could asked for help. Finally, Ore, the tester, were assigned all testing-related tasks.

All team members participated in the brainstorming, the video demonstration and the writing of the report. The team leader and the secretary wrote the project management report together.

6) Project Planning
6.1) Work Breakdown Structure

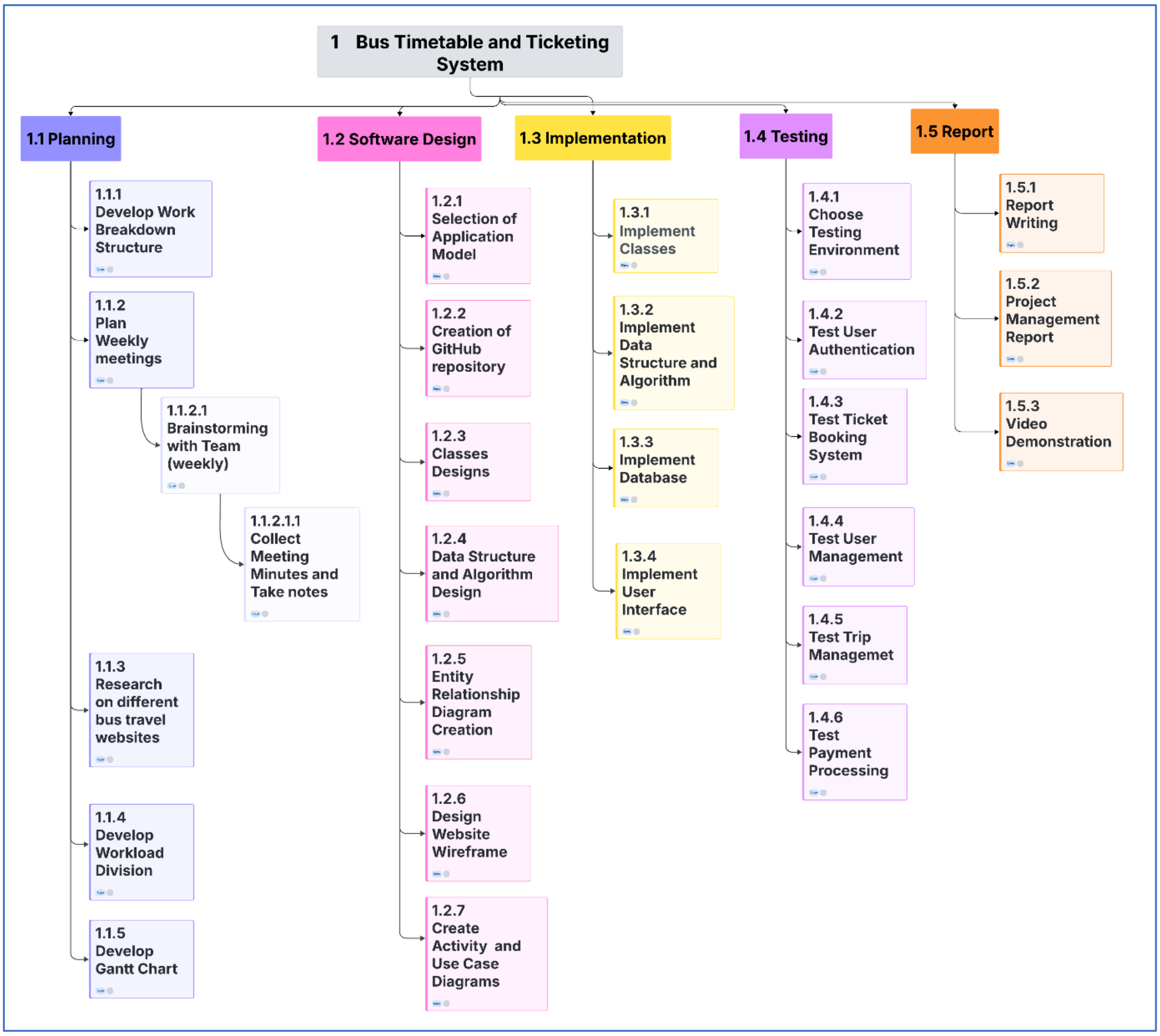

Figure 2 : Work Breakdown Structure

In figure 2 above, we can see that the project was decomposed into five different stages: the planning, the software design, the implementation, the testing, and the report. All tasks were categorized in one of these stages. The Work Breakdown Structure helped us with our organization; all the tasks of each stage; except those including the weekly meetings, brainstorming and collecting of meeting minutes, which needed to be done weekly during the entire duration of the project; had to be done before the beginning of the next stage. For example the tasks in the Planning stage had to be done before the tasks in the Software Design stage.

6.2) Gantt Chart

Project Name:	Bus Timetable and Ticketing System
Project Start	19/01/2026
Current Date	13/04/2026
Weeks in Progress	Wk 12
Team Members	Leo, Terry, Anastasia, Ivan, Ore

Tasks	Assigned Team Member(s)	Start Date	End Date	Days	Progress
Planning Stage					
Develop Work Breakdown Structure	Leo D'Hotman de Villiers	19/01/2026	23/01/2026	4	100%
Plan Weekly Meetings	Terry Leung Chew	19/01/2026	08/04/2026	79	100%
Brainstorming with Team (weekly)	Everyone	21/01/2026	08/04/2026	77	100%
Collect Meeting Minutes and Take notes	Terry Leung Chew	19/01/2026	08/04/2026	79	100%
Research on Different bus travel websites' designs	Anastasia Bholee and Ivan Pavlovskii	19/01/2026	02/02/2026	14	100%
Develop Workload Division	Leo D'Hotman de Villiers	23/01/2026	24/01/2026	1	100%
Develop Gantt Chart	Leo D'Hotman de Villiers	26/01/2026	27/01/2026	1	100%
Software Design Stage					
Selection of Application Model	Anastasia Bholee and Ivan Pavlovskii	03/02/2026	04/02/2026	1	100%
Creation of GitHub Repository	Leo D'Hotman de Villiers	13/02/2026	14/02/2026	1	100%
Classes Designs	Anastasia Bholee and Ivan Pavlovskii	16/02/2026	02/03/2026	14	100%
Data Structure and Algorithm Design	Anastasia Bholee and Ivan Pavlovskii	22/02/2026	13/03/2026	19	100%
Entity Relationship Diagram Creation	Leo D'Hotman de Villiers	23/02/2026	02/03/2026	7	100%
Design Website Wireframe	Anastasia Bholee and Ivan Pavlovskii	03/03/2026	09/03/2026	6	100%
Create Activity and Use Case Diagrams	Leo D'Hotman de Villiers	06/03/2026	10/03/2026	4	100%
Implementation Stage					
Implement Classes	Anastasia Bholee and Ivan Pavlovskii	07/03/2026	16/03/2026	9	100%
Implement Data Structure and Algorithm	Anastasia Bholee and Ivan Pavlovskii	07/03/2026	30/03/2026	23	100%
Implement Database	Anastasia Bholee and Ivan Pavlovskii	13/03/2026	31/03/2026	18	100%
Implement User Interface	Anastasia Bholee and Ivan Pavlovskii	13/03/2026	01/04/2026	19	100%
Testing Stage					
Choose Testing Environment	Ore Orekoya	16/03/2026	18/03/2026	2	100%
Test User Authentication	Ore Orekoya	19/03/2026	21/03/2026	2	100%
Test Ticket Booking System	Ore Orekoya	22/03/2026	24/03/2026	2	100%
Test User Management	Ore Orekoya	25/03/2026	30/03/2026	5	100%
Test Trip Management	Ore Orekoya	31/03/2026	02/04/2026	2	100%
Test Payment Processing	Ore Orekoya	31/03/2026	04/04/2026	4	100%
Report Stage					
Report Writing	Everyone	18/03/2026	12/04/2026	25	100%
Project Management Report	Leo D'Hotman de Villiers and Terry Leung Chew	18/03/2026	15/04/2026	28	100%
Video Demonstration	Everyone	30/03/2026	08/04/2026	9	100%

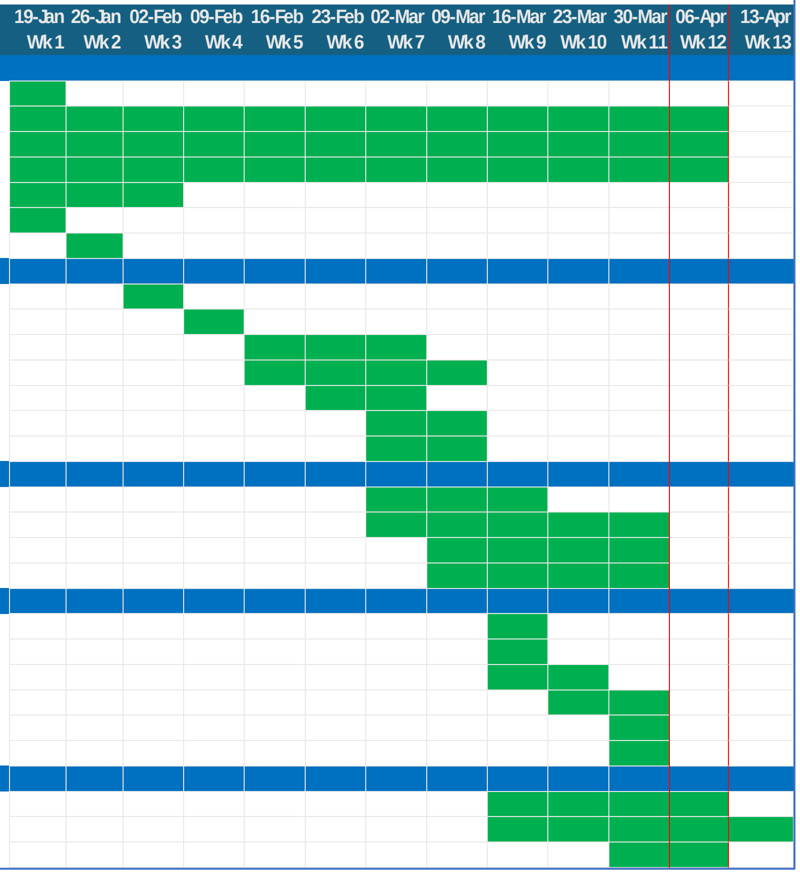

Figure 3: Gantt Chart

Figure 3, above, shows us the timeline of our project. This timeline was created with all the tasks that are present in the Work Breakdown Structure, taking into account the hierarchy of the different stages, for example, the fact that the tasks that were in the planning stage needed to be done before the tasks in the software design stage.

If ever a task needed more time to be completed, the team member responsible for the task would send a message to the WhatsApp group, and the end date of the corresponding task was postponed to a later date.

The deadline of the project was originally the 10th of April, but it changed and was moved to the 16th of April, so a week was added into the Gantt Chart, and the end date of the remaining tasks, the report, and the project management report that were nearly done, was postponed to a later date. The additional time was used to improve these two reports.

7) Project Design and Implementation Management

The design decisions were discussed during our weekly meetings. In the beginning the team only wanted to include a path in our website for a default user . But then the admin role and moderator role were created and implemented. To support the development three activity diagrams were created, one for each role.

7.1) Activity Diagrams

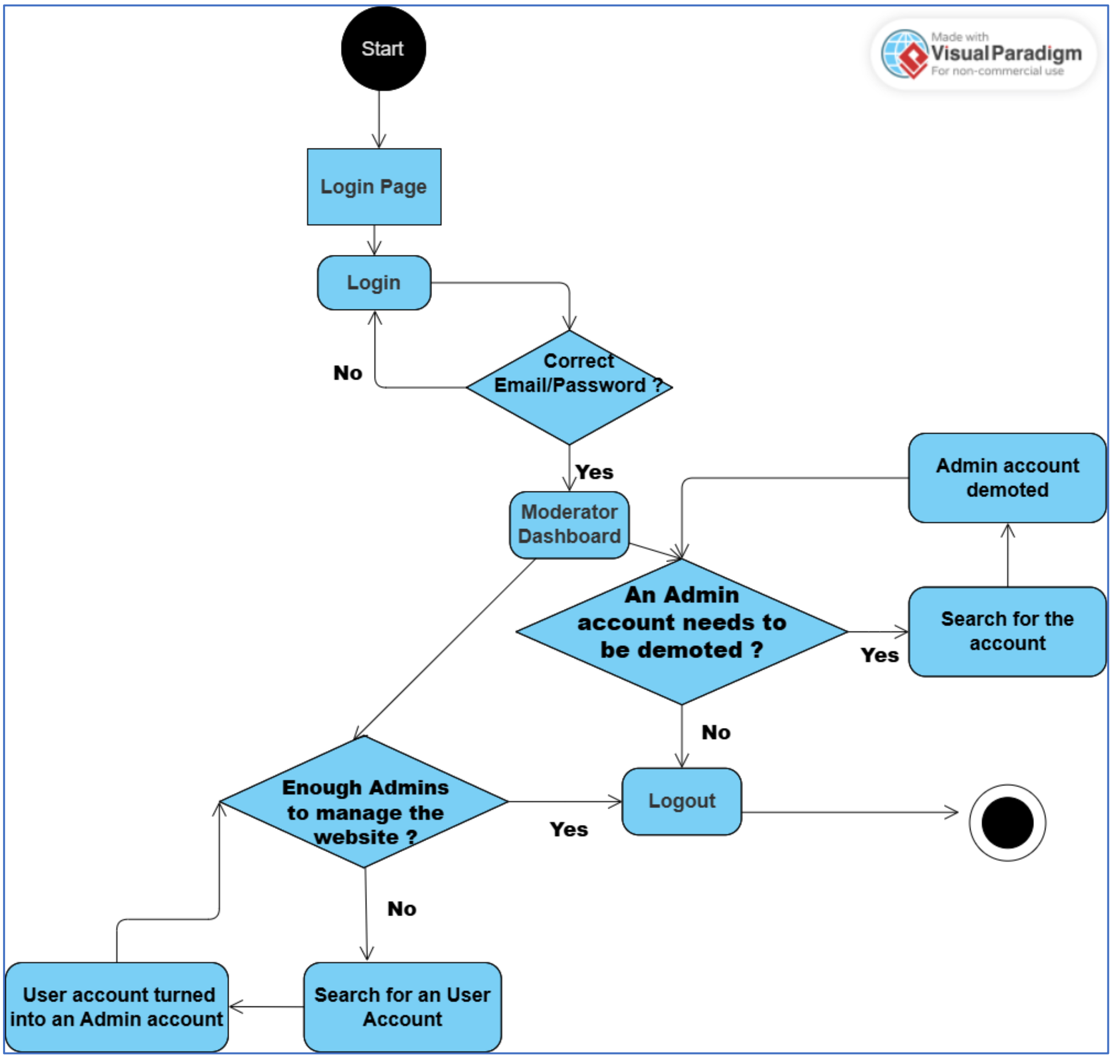

Figure 4 : Moderator Activity Diagram

Figure 4 above shows us that the moderator role consisted of giving the role of admin to the default user or demoting admin to default use

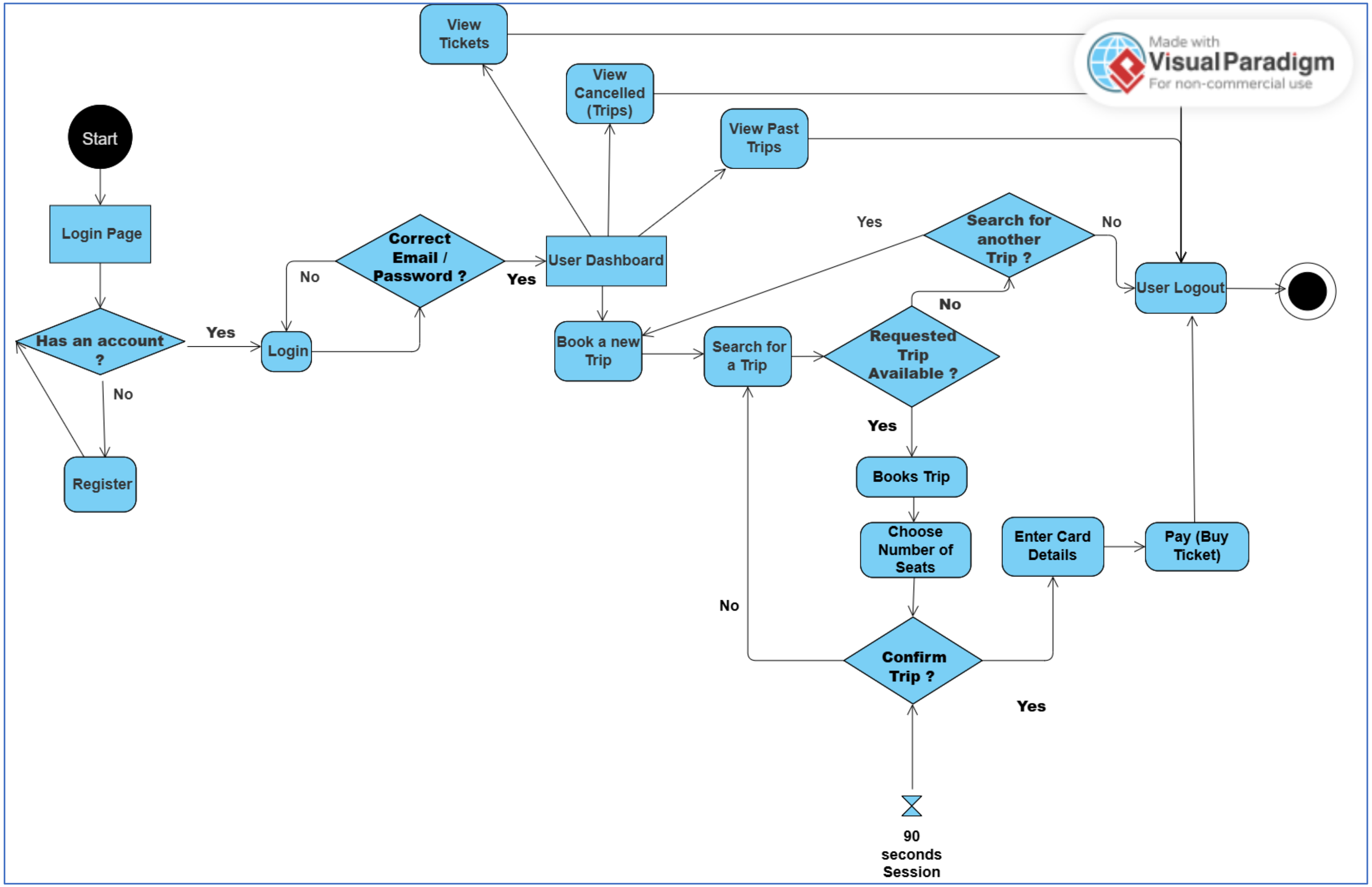

Figure 5 : User Activity Diagram

Figure 5 above shows us that the default user role consisted of searching for a trip in our bus timetable, selecting it, and buying a ticket for the corresponding trip. A default user can also see his past trips, his tickets and the trips cancelled by an admin

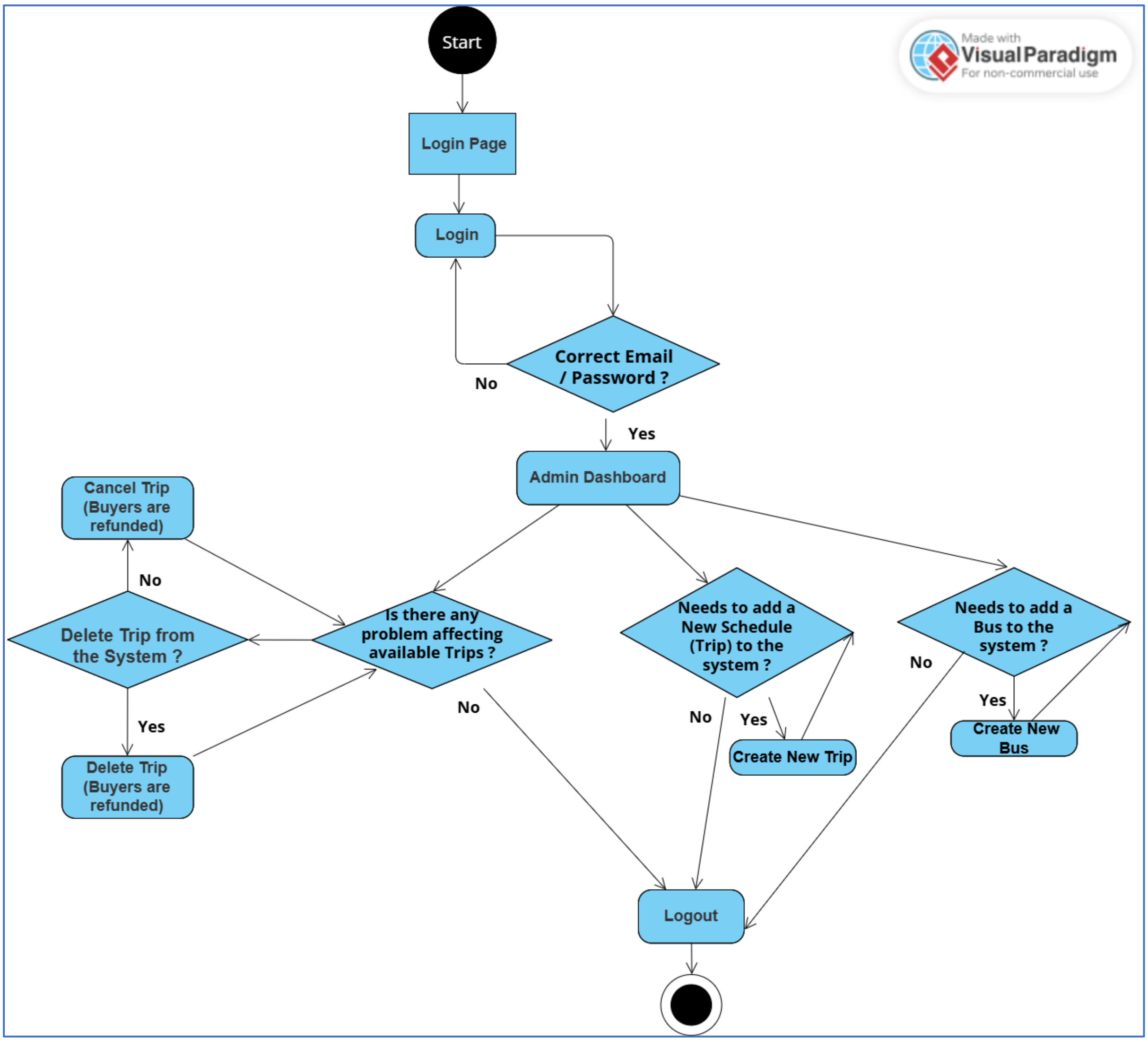

Figure 6 : Admin Activity Diagram

Figure 6 shows above shows us that the admin role consisted of deleting / cancelling trips when needed, as well as adding new trips or new buses to the bus timetable.

7.2) Use Case Diagram

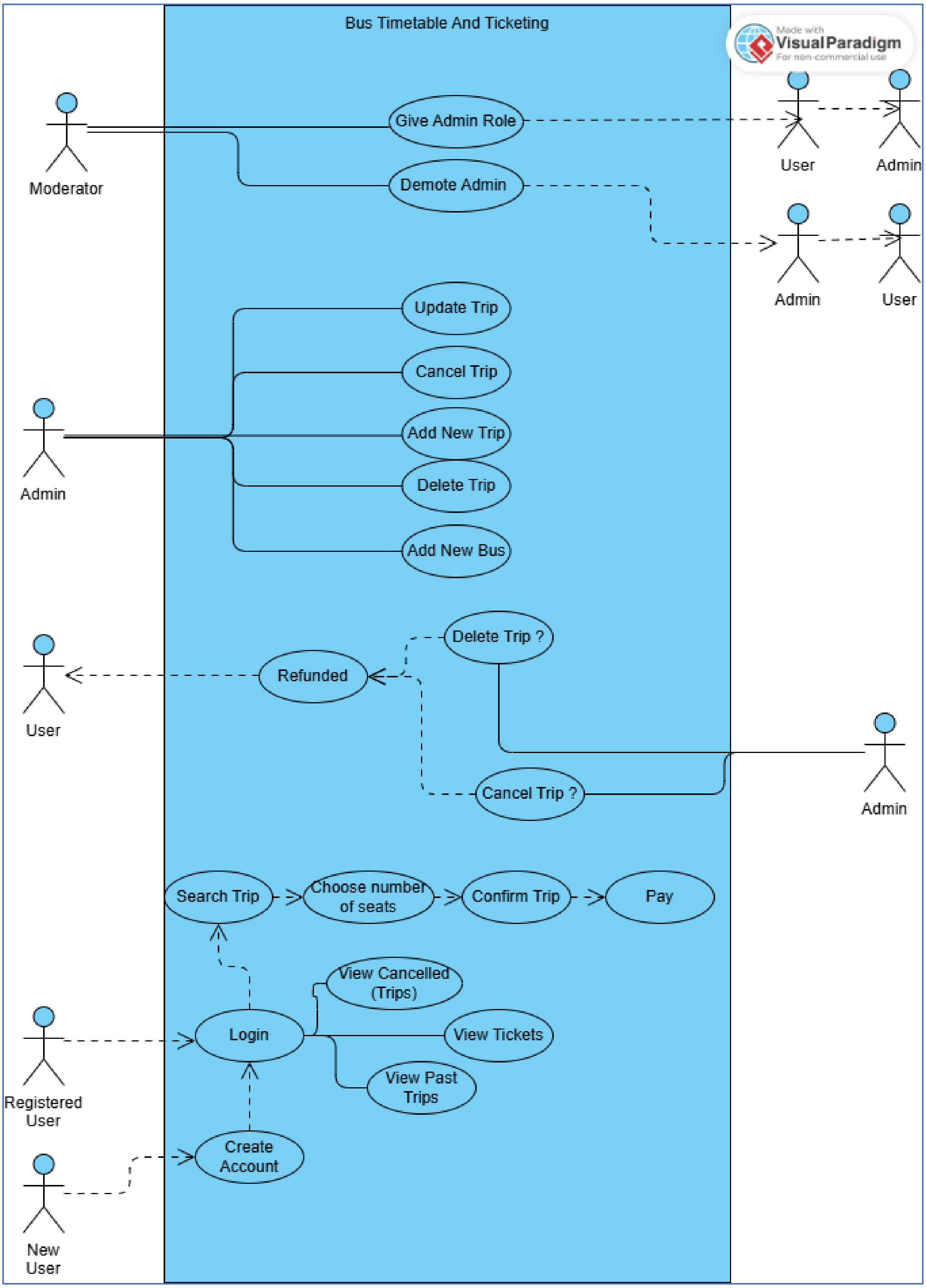

Figure 7 : Use Case Diagram

Figure 7 above shows us the different functions done by the different actors in the system Bus Timetable and Ticketing.

7.3 Website Wireframes

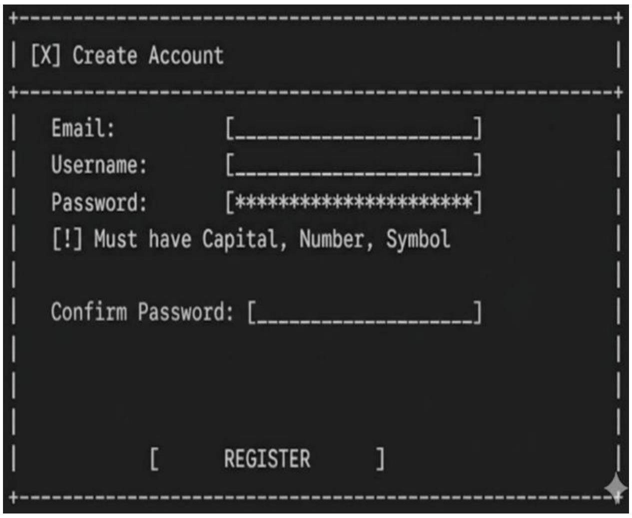

Figure 8: Login Form Wireframe

Figure 9: Register Form Wireframe

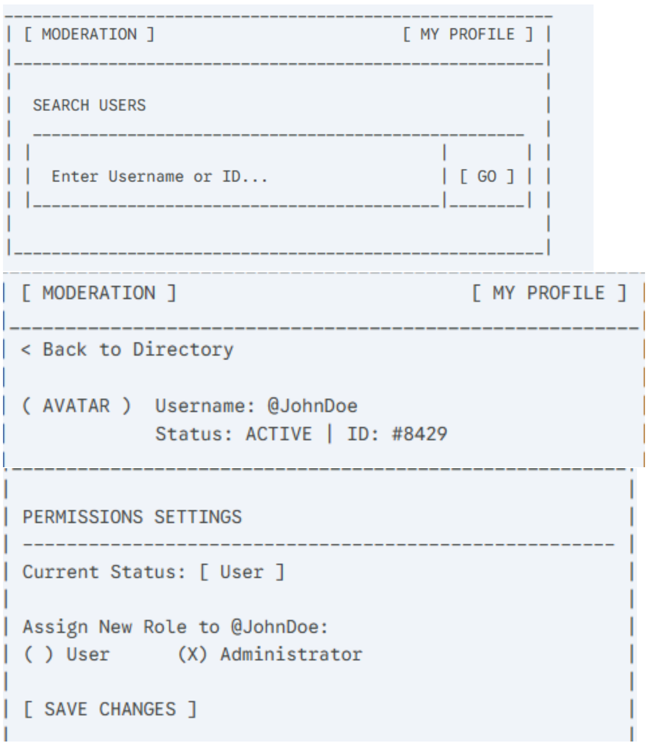

Figure 10 Moderator Dashboard Wireframe

```
+----------------------------------------------------------+
| [X] Admin Control Panel - CityTransit                    |
+----------------------------------------------------------+
|  [ View Bookings ] [ Manage Schedules ] [ Logout ]       |
|----------------------------------------------------------|
|  ADD NEW SCHEDULE:                                       |
|  From: [________]  To: [________]  Date: [10/03/26]      |
|  Dep:  [08:00   ]  Arr: [11:30   ]  Price:[ 20.00 ]      |
|                  [ ADD SCHEDULE ]                        |
|----------------------------------------------------------|
|  CURRENT SCHEDULES:                                      |
|  +----------------------------------------------------+  |
|  | Route          | Date  | Bookings | Action         |  |
|  |----------------|-------|----------|----------------|  |
|  | Ham -> Ber     | 3 Mar |    12    | [Cancel Trip]  |  |
|  | Ber -> Ham     | 4 Mar |     0    | [Delete]       |  |
|  +----------------------------------------------------+  |
+----------------------------------------------------------+
```

Figure 11: Admin Dashboard Wireframe

```
+--------------------------------------------------------------------------+
| [X] CatanBus - User Dashboard | Welcome, [John Doe]                      |
+--------------------------------------------------------------------------+
|  ( BUS LOGO )  CatanBus          [ VIEW ALL TRIPS ] [ VIEW MY TICKETS ]
|--------------------------------------------------------------------------|
|  Search Bar System                                                       |
|  From Location:             To Location:            Date:    [SEARCH]    |
|  [ Hamburg      v ]         [ Berlin       v ]      [11/12/2026]         |
|  Passengers: [ 2 Adults v ]                                              |
|--------------------------------------------------------------------------|
| +-------------------+--------------------+-------------------------+     |
| | Book a New Trip   |   My Past Trips    |   My Cancelled Logs     |     |
| +-------------------+--------------------+-------------------------+     |
```

Figure 12: User Dashboard Wireframe

Figures 8 to 12 above are examples of design wireframes we created before starting the code. They are not terminal outputs.
These wireframes helped us to have a better idea of how we could design the future forms and supported the development.

8) Testing Management

The person responsible for the testing was Ore. He used the MSTest framework to performed his tests, he worked closely with the two developers of our team which would tell him which part of the code needed to be test.

The bugs were reported either directly by message to the developers, or in the WhatsApp group, or during the weekly meetings.

The main part of the testing were done after the main part of the code was finished.

9) Risk Management

The different risks we had were:

- A team coordination risk, where the developers had some problems with the implementation of the different functionalities. What caused it was miscommunication inside our team. To resolve this issue, we discussed the different functionalities that needed to be implemented during our weekly meeting, each team member proposed different ideas; the idea of having different roles came from one of these brainstorming sessions. The team learned to better communicate and to ask for help when needed.

- A time-related risk, where in the beginning of the project we did not know how to manage our time. What helped was the creation of the Gantt Chart. We learned to manage our time by allocating a start date and an end date to our tasks.

- Another team coordination risk, where two team members worked on the same part of the report. This issue was mitigated by taking ideas from both different versions and merging them. What was learned was to allocate different part of the report to different team member.